Tranexamic Acid

Mechanism of action

- Tranexamic acid is an antifibrinolytic medicine. It blocks the conversion of plasminogen to plasmin, reducing fibrinolysis (breakdown of blood clots) and bleeding.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Postpartum haemorrhage.
- ✓ Clinically significant bleeding or signs of hypovolaemia following trauma.
- ✓ Non-traumatic bleeding and shock is severe.
- ✓ Cardiac arrest secondary to trauma.
- ✓ Bleeding following tonsillectomy.
- ✓ Crush injury.
- ✓ Moderate to severe epistaxis, not controlled following one dose of adrenaline IN.
- ✓ Antepartum haemorrhage if shock and/or bleeding is severe.

Contraindications

- ✕ Known severe allergy.
- ✕ Trauma when tranexamic acid will be administered more than three hours after the time of injury. This appears to increase mortality rates, the cause of which is not clear.

Cautions

- ⚠ None.

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Use in pregnancy or when breastfeeding

- Safety has not been demonstrated, but the balance of risk is such that it should be administered if indicated.

Dosage

- For bleeding following tonsillectomy:
  - 500 mg nebulised for an adult or child. Repeat once after 30 minutes.
  - 1 g IV for an adult. See the paediatric drug dose tables for a child.
- For bleeding following trauma:
  - 2 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- Non-traumatic bleeding:
  - 1 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- For moderate to severe epistaxis:
  - Administer 200 mg (2 ml undiluted) per bleeding nostril for a patient aged 12 years or over.
  - Administer 100 mg (1 ml undiluted) per bleeding nostril for a child aged 5-11 years.
  - Seek clinical advice if the patient is a child aged less than five years.

Administration

- IV: administer over 1-2 minutes.
- Nebulised: administer undiluted.
- IN: administer undiluted.

Common adverse effects

- None.

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Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 8 hours.

Usual preparation

- 10 ml ampoule containing 1 g.
- 5 ml ampoule containing 500 mg.

Pharmacokinetics

- Tranexamic acid is predominantly excreted in the urine.
- Renal impairment does not alter the initial (loading) dose.

Common interactions

- None.

Additional information

- Some references describe the topical administration of tranexamic acid for control of superficial bleeding. This usually has no role in the out-of-hospital setting and personnel should seek clinical advice if it is being considered.

More Information

Version history

9 June 2025 1.0.5.3
Update: Indications and dosage updated.

28 October 2024 1.0.5
Update: Indications, dosage and route updated.

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11 September 2023 1.0.4

Update: Indications have been updated. Reference to PATCH trial removed.